

Problem 8.16. The game of *Nim* is played with a collection of piles of sticks. In one move, a player may remove any nonzero number of sticks from a single pile. The players alternately take turns making moves. The player who removes the very last stick loses. Say that we have a game position in Nim with k piles containing $s_1, \dots, s_k$ sticks. Call the position balanced if each column of bits contains an even number of 1s when each of the numbers s_i is written in binary, and the binary numbers are written as rows of a matrix aligned at the low order bits. Prove the following two facts.

Part a. Starting in an unbalanced position, a single move exists that changes the position into a balanced one.

Proof. □

Part b. Starting in a balanced position, every single move changes the position into an unbalanced one.

Proof. □

Part c. Let $NIM = \{\langle s_1, \dots, s_k \rangle \mid \text{each } s_i \text{ is a binary number and Player I has a winning strategy in the Nim game starting at this position}\}$. Use the preceding facts about balanced positions to show that $NIM \in L$.

Proof. □